

SYNOPSIS: PIZZA SALES ANALYSIS DASHBOARD USING DASH

1. Introduction

In today's data-driven business environment, analyzing sales performance is crucial for making informed decisions. This project focuses on developing an **interactive Pizza Sales Dashboard** using Python libraries such as Dash and Plotly to visualize key performance indicators (KPIs) and identify business trends. The system provides a comprehensive view of sales, customer preferences, and product performance.

2. Objective

The primary objective of this project is to:

- Analyze pizza sales data to extract meaningful insights
- Monitor business performance using key KPIs
- Identify customer preferences and sales trends
- Enable interactive data exploration through dynamic dashboards

3. Key Performance Indicators (KPIs)

The dashboard evaluates the following KPIs:

- **Total Revenue:** Total income generated from all pizza sales
 - **Average Order Value:** Average amount spent per order
 - **Total Pizzas Sold:** Total quantity of pizzas sold
 - **Total Orders:** Total number of unique orders placed
 - **Average Pizzas per Order:** Average number of pizzas per order
- These KPIs help assess overall business performance and efficiency.

4. Data Visualization & Charts

To gain deeper insights, multiple visualizations are implemented:

- **Daily Trend (Bar Chart):** Shows variation in order volume across days of the week
- **Monthly Trend (Line Chart):** Identifies peak months and seasonal demand patterns
- **Sales by Category (Pie Chart):** Highlights contribution of each pizza category
- **Sales by Size (Pie Chart):** Analyzes customer preference for pizza sizes
- **Funnel Chart (Category-wise Sales):** Compares total pizzas sold across categories
- **Top 5 Best Sellers:** Displays highest performing pizzas based on revenue, quantity, and orders
- **Bottom 5 Worst Sellers:** Identifies underperforming pizzas

5. Methodology

The project follows these steps:

1. **Data Collection:** Importing pizza sales dataset using Pandas
2. **Data Preprocessing:**

- Converting date formats
 - Extracting day and month features
- 3. **Data Analysis:**
 - Aggregating data using groupby functions
 - Calculating KPIs
- 4. **Visualization:**
 - Creating interactive charts using Plotly
- 5. **Dashboard Development:**
 - Building an interactive UI using Dash
 - Implementing filters using callback functions

6. Tools & Technologies Used

- **Programming Language:** Python
- **Libraries:** Pandas, Plotly, Dash
- **IDE:** VS Code / Jupyter Notebook
- **Visualization:** Plotly Express
- **Framework:** Dash (for interactive web dashboard)

7. Features of the Dashboard

- Interactive filtering (day, month, category, size)
- Real-time KPI updates
- User-friendly interface
- Dynamic and responsive charts
- Clear identification of trends and patterns

8. Results & Insights

The dashboard enables:

- Identification of peak sales days and months
- Understanding of most popular pizza categories and sizes
- Recognition of top-performing and underperforming products
- Better decision-making for inventory and marketing strategies

Code Implementation:

```
import dash

from dash import dcc, html

from dash.dependencies import Input, Output

import pandas as pd

import plotly.express as px

# Load dataset
```

```
df = pd.read_csv(r"C:\Users\Kavyashree N\OneDrive\Desktop\Data-Science-Internship\week
7\pizza_sales\pizza_sales.csv")
```

```
# Data preprocessing
```

```
df['order_date'] = pd.to_datetime(df['order_date'], format='%d-%m-%Y')
```

```
df['month'] = df['order_date'].dt.strftime('%b')
```

```
df['day'] = df['order_date'].dt.day_name()
```

```
month_order = ['Jan','Feb','Mar','Apr','May','Jun',
               'Jul','Aug','Sep','Oct','Nov','Dec']
```

```
day_order = ['Monday','Tuesday','Wednesday','Thursday',
             'Friday','Saturday','Sunday']
```

```
app = dash.Dash(__name__)
```

```
app.layout = html.Div([
```

```
    html.H1(" 🍕 Pizza Sales Interactive Dashboard", style={'textAlign':'center'}),
```

```
    html.Br(),
```

```
# KPI Section
```

```
html.Div([
```

```
    html.Div(id='revenue'),
```

```
    html.Div(id='orders'),
```

```
    html.Div(id='pizzas'),
```

```
    html.Div(id='avg_order'),
```

```
    html.Div(id='avg_pizza')
```

```
], style={'display':'flex','justifyContent':'space-around','fontSize':'20px'}),
```

```
html.Br(),
```

```
# Trends
```

```
html.Div([  
    dcc.Graph(id='daily_chart'),  
    dcc.Graph(id='monthly_chart')  
], style={'display':'flex'}),
```

```
# Breakdown
```

```
html.Div([  
    dcc.Graph(id='category_chart'),  
    dcc.Graph(id='size_chart')  
], style={'display':'flex'}),
```

```
html.Br(),
```

```
# Funnel
```

```
dcc.Graph(id='funnel_chart'),
```

```
html.H3("🏆 Top 5 Best Selling Pizzas"),
```

```
html.Div([  
    dcc.Graph(id='top_revenue'),  
    dcc.Graph(id='top_quantity'),  
    dcc.Graph(id='top_orders')  
], style={'display':'flex'}),
```

```
html.H3("💀 Bottom 5 Worst Selling Pizzas"),
```

```
html.Div([  
    dcc.Graph(id='worst_revenue'),
```

```
        dcc.Graph(id='worst_quantity'),
        dcc.Graph(id='worst_orders')
    ], style={'display':'flex'})

])
```

```
@app.callback(
    [
        Output('revenue','children'),
        Output('orders','children'),
        Output('pizzas','children'),
        Output('avg_order','children'),
        Output('avg_pizza','children'),

        Output('daily_chart','figure'),
        Output('monthly_chart','figure'),

        Output('category_chart','figure'),
        Output('size_chart','figure'),

        Output('funnel_chart','figure'),

        Output('top_revenue','figure'),
        Output('top_quantity','figure'),
        Output('top_orders','figure'),

        Output('worst_revenue','figure'),
        Output('worst_quantity','figure'),
        Output('worst_orders','figure')
    ],
    [
```

```

    Input('daily_chart','clickData'),
    Input('monthly_chart','clickData'),
    Input('category_chart','clickData'),
    Input('size_chart','clickData')
]
)

def update_dashboard(day_click, month_click, category_click, size_click):

    filtered_df = df.copy()

    # Day filter
    if day_click:
        day = day_click['points'][0]['x']
        filtered_df = filtered_df[filtered_df['day'] == day]

    # Month filter
    if month_click:
        month = month_click['points'][0]['x']
        filtered_df = filtered_df[filtered_df['month'] == month]

    # Category filter
    if category_click:
        category = category_click['points'][0]['label']
        filtered_df = filtered_df[filtered_df['pizza_category'] == category]

    # Size filter
    if size_click:
        size = size_click['points'][0]['label']
        filtered_df = filtered_df[filtered_df['pizza_size'] == size]

    # KPI

```

```
revenue = filtered_df['total_price'].sum()
orders = filtered_df['order_id'].nunique()
pizzas = filtered_df['quantity'].sum()
avg_order = revenue / orders if orders else 0
avg_pizza = pizzas / orders if orders else 0
```

```
revenue_card = f"💰 Revenue: ${revenue:,.2f}"
orders_card = f"📦 Orders: {orders}"
pizzas_card = f"🍕 Pizzas Sold: {pizzas}"
avg_card = f"🇮🇹 Avg Order Value: ${avg_order:,.2f}"
avg_pizza_card = f"🍕 Avg Pizzas/Order: {avg_pizza:,.2f}"
```

```
# Charts
```

```
# Daily Trend
```

```
daily = filtered_df.groupby('day')['order_id'].nunique().reindex(day_order).reset_index()
fig_daily = px.bar(daily, x='day', y='order_id', title="Daily Orders")
```

```
# Monthly Trend
```

```
monthly = filtered_df.groupby('month')['order_id'].nunique().reset_index()
monthly['month'] = pd.Categorical(monthly['month'], categories=month_order, ordered=True)
monthly = monthly.sort_values('month')
fig_month = px.line(monthly, x='month', y='order_id', markers=True, title="Monthly Orders")
```

```
# Category Pie
```

```
category_sales = filtered_df.groupby('pizza_category')['total_price'].sum().reset_index()
fig_category = px.pie(category_sales, names='pizza_category', values='total_price', title="Sales by Category")
```

```
# Size Pie
```

```
size_sales = filtered_df.groupby('pizza_size')['total_price'].sum().reset_index()
```

```
fig_size = px.pie(size_sales, names='pizza_size', values='total_price', title="Sales by Size")
```

```
# Funnel Chart
```

```
funnel = filtered_df.groupby('pizza_category')['quantity'].sum().reset_index()
```

```
fig_funnel = px.funnel(funnel, x='quantity', y='pizza_category', title="Total Pizzas Sold by Category")
```

```
# Pizza Performance
```

```
pizza_perf = filtered_df.groupby('pizza_name').agg({
```

```
    'total_price': 'sum',
```

```
    'quantity': 'sum',
```

```
    'order_id': 'nunique'
```

```
}).reset_index()
```

```
# Top 5
```

```
top_rev = pizza_perf.sort_values('total_price', ascending=False).head(5)
```

```
top_qty = pizza_perf.sort_values('quantity', ascending=False).head(5)
```

```
top_ord = pizza_perf.sort_values('order_id', ascending=False).head(5)
```

```
fig_top_rev = px.bar(top_rev, x='total_price', y='pizza_name', orientation='h', title="Top Revenue")
```

```
fig_top_qty = px.bar(top_qty, x='quantity', y='pizza_name', orientation='h', title="Top Quantity")
```

```
fig_top_ord = px.bar(top_ord, x='order_id', y='pizza_name', orientation='h', title="Top Orders")
```

```
# Bottom 5
```

```
worst_rev = pizza_perf.sort_values('total_price').head(5)
```

```
worst_qty = pizza_perf.sort_values('quantity').head(5)
```

```
worst_ord = pizza_perf.sort_values('order_id').head(5)
```

```
fig_worst_rev = px.bar(worst_rev, x='total_price', y='pizza_name', orientation='h', title="Worst Revenue")
```



```
fig_worst_qty = px.bar(worst_qty, x='quantity', y='pizza_name', orientation='h', title="Worst
Quantity")
```

```
fig_worst_ord = px.bar(worst_ord, x='order_id', y='pizza_name', orientation='h', title="Worst
Orders")
```

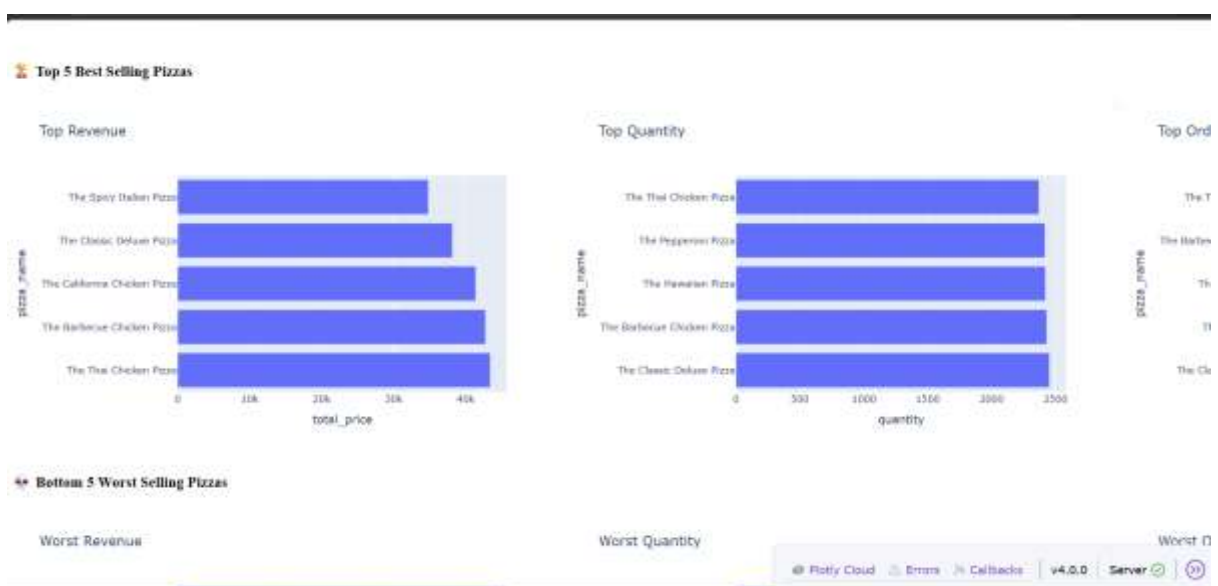
```
return (
    revenue_card, orders_card, pizzas_card, avg_card, avg_pizza_card,
    fig_daily, fig_month,
    fig_category, fig_size,
    fig_funnel,
    fig_top_rev, fig_top_qty, fig_top_ord,
    fig_worst_rev, fig_worst_qty, fig_worst_ord
)
```

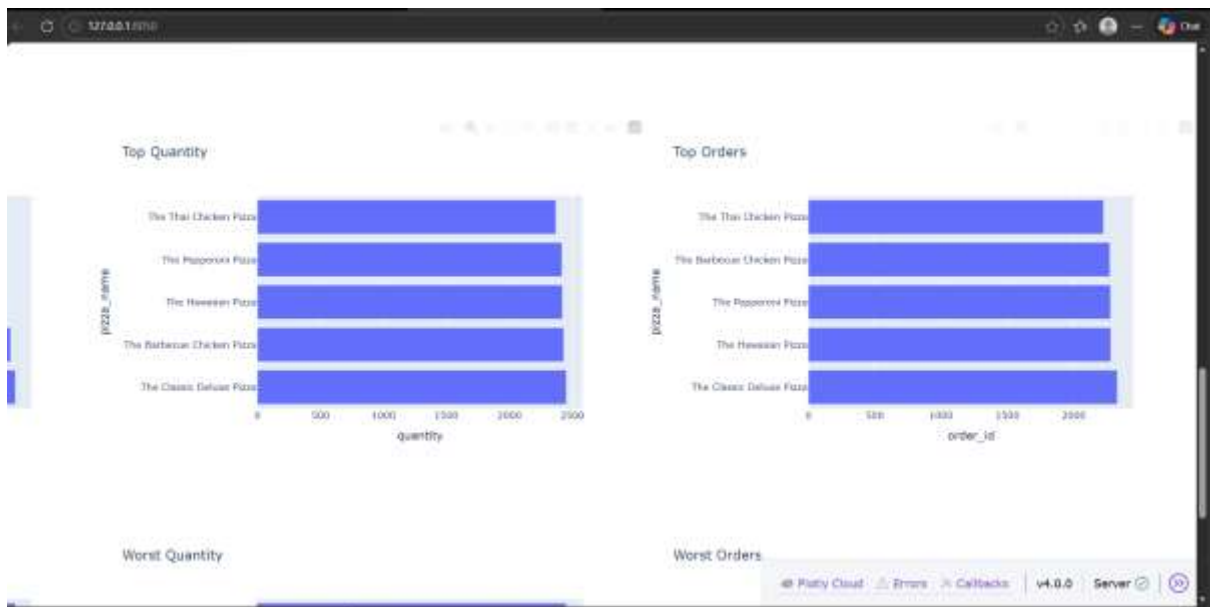
```
if __name__ == '__main__':
```

```
    app.run(debug=True)
```

Output:









9. Conclusion

This project demonstrates how **data visualization and analytics** can transform raw sales data into actionable insights. The interactive dashboard helps businesses monitor performance, understand customer behavior, and optimize operations effectively.